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Design of human-machine interaction testing tool for drilling instruments based on CAN bus

Qimengsha Zhang*, Yongjian Hu, Chengqin Sun, Qi Sun, Guanjie Zhang and Meng Xiao

Beijing Petroleum Machinery Co., Ltd, Beijing, 102206, China

*zhangqimengsha@163.com

Abstract: A small, portable, and user-friendly human-machine interaction testing tool for drilling instruments is designed and developed to accommodate the underground multi-parameter measurement environment and to meet the ground testing requirements of the drilling instruments. To improve the reliability of data transmission, we customized the CAN (Controller Area Network) application layer protocol by dividing the identifier and data frames reasonably. The designed prototype can send multiple commands to the drilling instruments, also monitoring CAN data flow and displays to users.

1. Introduction

With the continuous growth of global demand for oil and gas resources, most domestic oilfield development has entered the mid to late stage. The focus of oil and gas drilling and production has shifted from easily exploitable middle and shallow layers to complex oil reservoirs such as ultra deep well, thin oil layers, and heavy oil layers with higher extraction difficulties. In order to improve single well production, oil and gas recovery rate, and development efficiency, directional drilling technologies with horizontal wells, multi branch wells, and large displacement wells need to be continuously improved, which puts forward higher requirements for the measurement and transmission of engineering parameters and geological parameters while drilling[1,2]. Obtaining engineering parameters such as torque, bending moment, vibration, temperature, and pressure while drilling is beneficial for assessing downhole conditions, identifying risks, and improving well control safety; obtaining geological parameters such as natural gamma and resistivity while drilling is beneficial for optimizing wellbore trajectory and improving drilling encounter rate[3]. With the increasing difficulty of oil and gas field exploitation and the development of drilling measurement technology, more and more downhole parameters will need to be measured.Therefore, it is necessary to optimize the communication network of drilling instruments to ensure the safe and efficient transmission of underground data, in order to adapt to measurement systems with more parameters.

Nowadays, drilling instrument manufacturers do not have a unified drilling instrument bus, and the most widely used method is the RS485 bus data transmission. Compared to RS485, CAN bus has higher anti-interference, flexibility, and bus utilization. In the industry, the CAN protocol is widely acknowledged as a high-performance and reliable communication protocol, and it finds extensive applications in fields like automotive electronics, industrial control, mechanical manufacturing, aerospace, and coal mining[4,5]. The short distance transmission rate can reach 1Mb/s, and the bus topology has good extensibility. The drilling instrument needs to be tested and configured on site before it can be lowered into the well. Therefore, an intelligent serial port display that supports CAN communication can be used to achieve human-machine interaction. Users can directly operate the lightweight and compact touch screen graphical interface to set up drilling instruments and monitor CAN data flow. This article proposes a communication network for downhole drilling instruments based on the CAN bus and uses the USART HMI (Universal Synchronous/Asynchronous Receiver/Transmitter Human Machine Interface) intelligent serial port display with a built-in CAN module for configuration. The customized CAN protocol is used for encoding and decoding to achieve real-time display of measurement data for drilling instruments.

2. Intelligent serial port display

This project uses a 7-inch USART-HMI intelligent serial port display, which includes a CAN communication module and is connected to the drilling instrument through the CAN bus. Compared to large laptops, the small and portable touch screen is very suitable for application environments such as assembly workshops and on sites, making it easy for users to get started and provide a better user experience[6]. This serial port display contains 36 configuration controls such as text, buttons, lists, CAN protocol parsers, etc., which can design a various and user-friendly human-machine interface. The upper computer software provided by the manufacturer supports writing scripts using C language syntax. This serial port display is designed with multiple functions to meet the testing and setting requirements of the circuit module of the drilling instrument.

3. CAN communication system for drilling instruments

3.1. CAN bus physical layer design

CAN bus was originally developed as a serial communication technology to solve the data exchange problem between various electronic units in the automotive industry, with a multi-master network structure[7]. There are various types of CAN bus topologies, such as linear, star, ring, etc. Considering the spatial structure of the drill collar, this design adopts a linear structure, as shown in figure 1. A linear topology requires terminal resistors R to be connected at both ends of the bus, with a resistance value of 120 Ω. The CAN bus network uses differential signal technology at the physical layer, and the communication carrier is usually a twisted pair cable named CAN-H and CAN-L respectively[8]. The measurement module is used to collect, store, and send measurement data while drilling, such as annular pressure, annular temperature, near-bit gamma, torque, tension, etc. The communication module is used to receive measurement module data and achieve wireless transmission of underground data, such as cross screw wireless communication.

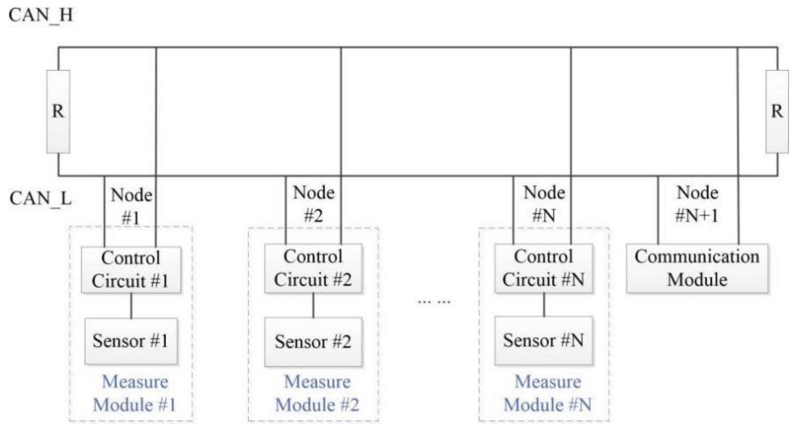


Figure 1. Linear topology.

Journal of Physics: Conference Series **2901** (2024) 012005 doi:10.1088/1742-6596/2901/1/012005

3.2. CAN protocol layer design

ISO (International Organization for Standardization) only defines the protocols for the physical layer and data link of the CAN bus in the 11898-1 standard, without specifying the application layer protocol[9].Therefore, users can customize the protocol according to different application scenarios. CAN frames are mainly divided into four types: data frames, remote control frames, error frames, and overload frames. Data frames are used for communication between intelligent serial port displays and drilling instruments. This article adopts the standard format of an 11 bit address ID (identifier), and the CAN bus standard data frame structure is shown in figure 2.

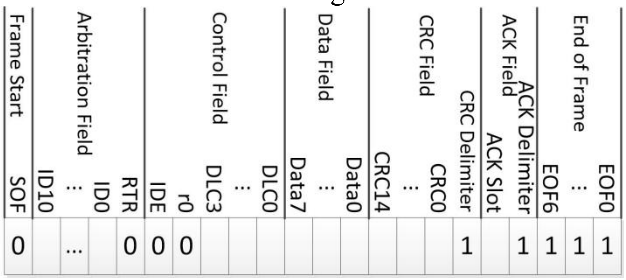


Figure 2. Standard data frame structure.

The division and definition of identifiers in the application layer protocol of CAN bus in this article are shown in table 1. The source node address represents the initiator's address and supports a maximum of 32 nodes. The target node address represents the address of the data receiver, supporting a maximum of 32 nodes. ID is used to indicate the priority of the arbitration segment of the data bus. The smaller the ID value, the higher the corresponding priority level and the later it is eliminated[10]. In this design, the addresses of "broadcast" and "PC" are set to have the highest priority, with ID values of "0" and "1" respectively. Other nodes are sorted according to the frequency of measurement data requirements. Broadcast can be sent to all nodes except oneself and receive broadcast messages sent by other nodes. "PC" is the address of the serial port display.

Table 1. ID defination.

ID10~ID6	ID5~ID1	ID0
Target node address	Source node address	–

The definition of data frames is shown in table 2. According to the CAN2.0 standard, data frames are defined as 8 bytes. Therefore, the frame number is used to represent the sequence of data frames, which is used to concatenate multiple data frames. If the frame number is 0, it indicates that complete data has been received and can be analyzed according to the application layer protocol. Byte 1 is divided into two parts: operation type and data type. This protocol shares 3 operation types as shown in table 3, and 6 data types as shown in table 4. Byte 2 represents the data length, byte 3 and the remaining bytes represent the data. If the data length is greater than 5 bytes, the message cannot be fully transmitted in an 8-byte data frame, the data will be broken down into multiple data packets. The data frame in the first packet is defined as shown in table 2, and the remaining data frames in the data packets only include data except for byte 0, which is the frame number.

Table 2. Data frame defination.

Byte 0	Byte 1		Byte 2	Byte 3~Byte7
	Bit7~Bit6	Bit5~Bit0		
Frame number	Operation type	Data type	Data length	Data

The CAN command operation types sent by the serial port display to the drilling instrument can be divided into three types: "read", "write", "write and read". By user selection, the read operation can display module return data on the corresponding control of the serial port display, including module

Journal of Physics: Conference Series **2901** (2024) 012005 doi:10.1088/1742-6596/2901/1/012005

specific information and measurement data. The write operation, for example, can make modules enter a designated self-check program. Write and read operations can be used to send controlling commands to the module and read returning data to confirm that the module has completed the designated operation.

Table 3. Operation type.

Operation type	Values
Read	0
Write	1
Write and read	2

The factory information data type is used when users require the date of manufacture and serial number of the instrument. The used time data type is used when users require the accumulative using time of the instrument. Data flow type mainly includes measurement data sent by various modules attached to the CAN bus, control instructions from the serial port display to each measurement module, and system initialization information. CAN command of self checking data type can control each module to enter different self checking programs, used to test the FRAM (ferroelectric memory), RTC (real-time clock), CAN communication of each module, and the wireless communication function of the communication module. The reset record is used to clear the measurement module's memory storage data and download the RTC time. The data types of tool face high side calibration include gravity tool face high side and magnetic tool face high side. Users can set the angle deviation through the serial port display measurement module to calibrate instruments, such as gamma sensors and directional sensors.The data for operation types “write” and “write and read” are the same. The operation type can be set according to the needs. For example, if it is necessary to confirm whether the instruction is successful after writing, the “write and read” operation can be used.

Table 4. Command type.

Data type	Operation type	Data
Factory information	Read	–
	Write	Date of manufacture and serial number
Used time	Read	–
	Write	Minute
Data flow	Read	Data
	Write	Date flow type and data
Self-check	Write	Index of the self-check program
Reset Record	Write	RTC
Tool face (TF) high side	Write	TF type

4. Serial port display design

4.1. Software design

In this design, the serial port display includes two major data processing functions: sending data to the drilling instrument in protocol format according to the touch interface operation; receiving drilling instrument data,and display it in the corresponding control after protocol parsing. The serial port display needs to monitor the data sent by all drilling instrument modules connected to the CAN, so there is no need to set a CAN filter. As shown in figure 3, the user's touch operation on the serial port display interface can trigger the data sending script. Based on the different functions of each control, the customized data frame will be sent to the target CAN module through the serial port display. As shown in figure 4, the serial port display triggers the CAN protocol parser data reception script every time it receives a frame of data. After protocol parsing, the data is displayed on the text control or log control.

Journal of Physics: Conference Series **2901** (2024) 012005 doi:10.1088/1742-6596/2901/1/012005

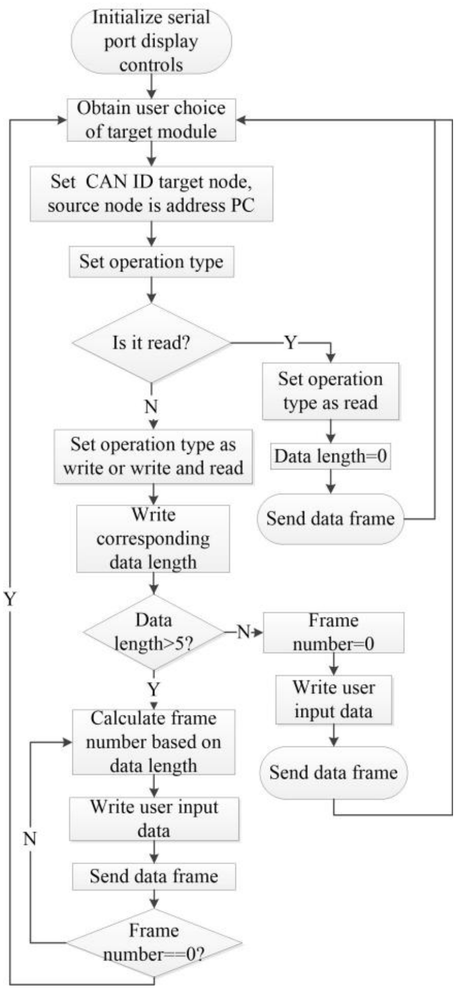


Figure 3. Data sending flow chart.

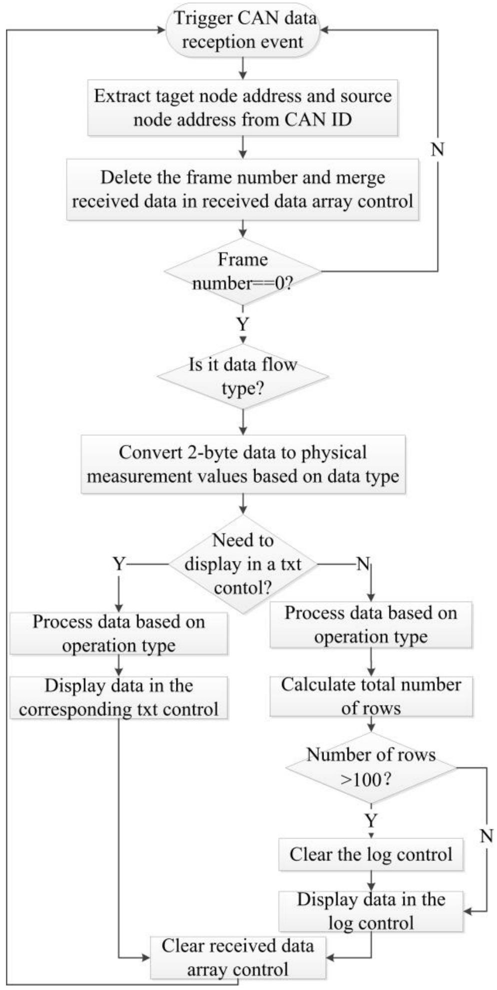


Figure 4. Data reception flow chart.

4.2. Interface design

The serial port display interface is divided into two areas. There is a log control in the monitoring area that can display data line by line. When the monitoring area is full of data, it scrolls up automatically. Once 100 lines of data are displayed, the log control will be automatically cleared. The data in the monitoring area can also be cleared by pressing the clear button in the operation area. The operation area allows users to send instructions to the module by clicking on corresponding buttons, as shown in figure 5.

Journal of Physics: Conference Series **2901** (2024) 012005 doi:10.1088/1742-6596/2901/1/012005

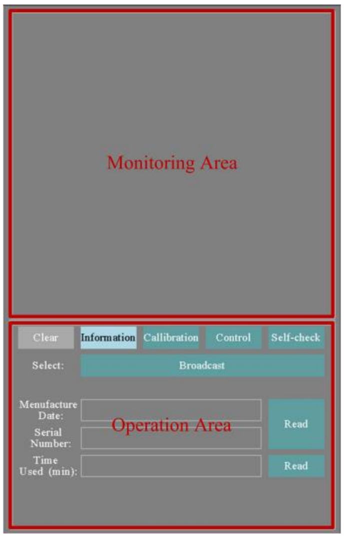


Figure 5. Monitor area and operation area.

According to the instrument's configuring and testing requirements, 4 pages are designed which are information, calibration, control, and self-check. The monitoring area on all four pages remains unified, and users can observe the data flow in the monitoring areas on any page. The operation area on each page is different but contains some identical controls: page switch controls and the selection button control, as shown in figure 6. After clicking the selection button control, a list of operation objects will pop up, each corresponding to a target node address. If "broadcast" is selected, all objects in the CAN network will execute the instructions sent by the serial port display.

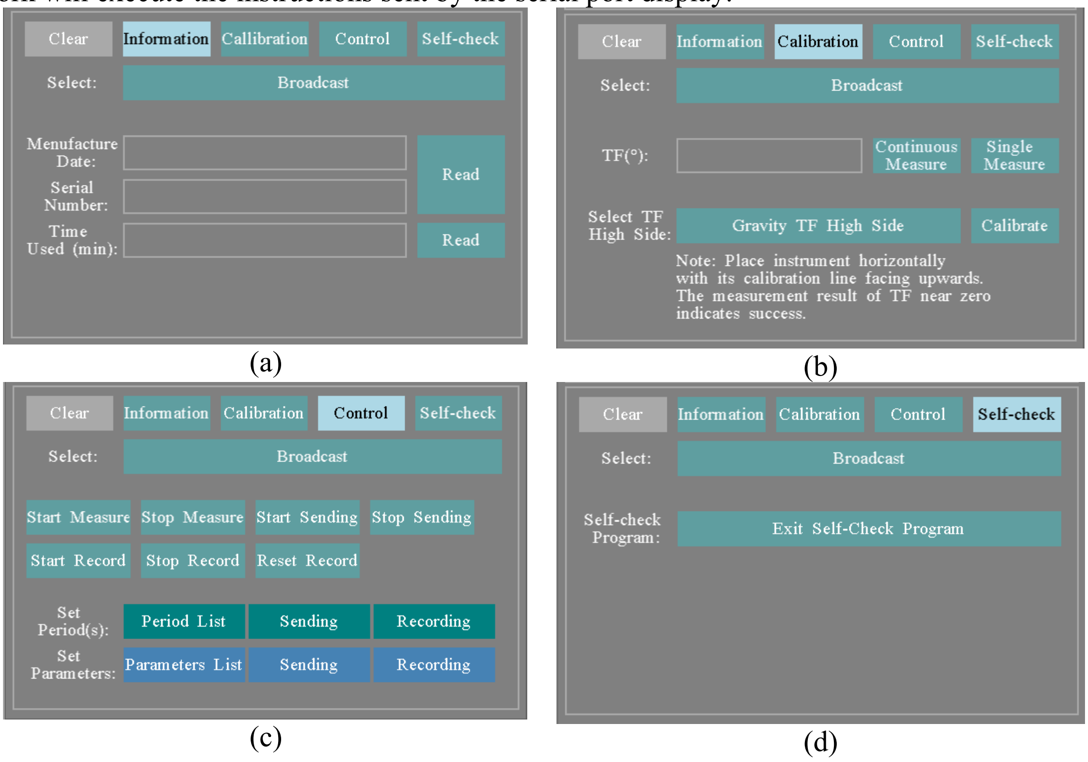


Figure 6. Design of operation area.

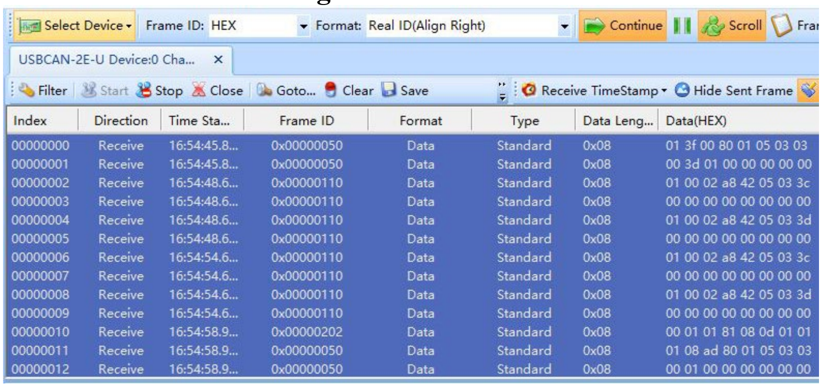
Journal of Physics: Conference Series **2901** (2024) 012005 doi:10.1088/1742-6596/2901/1/012005

5. Physical testing

The serial port display is connected to the upper computer through a USB interface, and the project is downloaded to the serial port display using the upper computer software provided by the manufacturer. To test the functions of the instrument testing tool, the serial port display is connected to the pressure and temperature while drilling (PTWD) module through the CAN bus. The monitoring area displays data flow information between each module, such as the source node (PTWD module), target node (NFC module), sending time, and measurement values. The monitoring area also displays system initialization information, measurement control information, and self-check program return values as shown in figure 7. During testing, connect the CAN bus professional analysis and testing tool to the CAN network, and use CAN bus analyser to monitor the data exchange between the serial port display and the PTWD module, as shown in figure 8. Testing shows that the serial port display can send commands and analyze the received data based on the designed application layer protocol and display the parsed CAN message correctly on corresponding controls.

16:54:45 PTWD→PC:Infomation code:System initialization:Successful
16:54:48 PTWD→NFC:Annular pressure:-0.10Mpa
16:54:48 PTWD→NFC:Annular temperature:-50°C
16:54:54 PTWD→NFC:Annular pressure:-0.10Mpa
16:54:54 PTWD→NFC:Annular temperature:-50°C
16:54:58 PTWD→PC:Infomation code:Enter self-check:Core
16:54:58 PTWD→PC:Circuit temperature:31.00°C
16:54:58 PTWD→PC:Inclination:1.00°
16:54:59 PTWD→PC:Inclination:2.00°
16:55:0 PTWD→PC:Inclination:3.00°
16:55:1 PTWD→PC:Inclination:4.00°
16:55:2 PTWD→PC:Information code:Exit self-check
16:55:7 PTWD→NFC:Annular pressure:-0.10Mpa
16:55:7 PTWD→NFC:Annular temperature:-50°C
16:55:10 PTWD→PC:Measurement control:Stop measure

Figure 7. Data flow.



Index	Direction	Time Sta...	Frame ID	Format	Type	Data Leng...	Data(HEX)
00000000	Receive	16:54:45.8...	0x00000050	Data	Standard	0x08	01 3f 00 80 01 05 03 03
00000001	Receive	16:54:45.8...	0x00000050	Data	Standard	0x08	00 3d 01 00 00 00 00 00
00000002	Receive	16:54:48.6...	0x00000110	Data	Standard	0x08	01 00 02 a8 42 05 03 3c
00000003	Receive	16:54:48.6...	0x00000110	Data	Standard	0x08	00 00 00 00 00 00 00 00
00000004	Receive	16:54:48.6...	0x00000110	Data	Standard	0x08	01 00 02 a8 42 05 03 3d
00000005	Receive	16:54:48.6...	0x00000110	Data	Standard	0x08	00 00 00 00 00 00 00 00
00000006	Receive	16:54:54.6...	0x00000110	Data	Standard	0x08	01 00 02 a8 42 05 03 3c
00000007	Receive	16:54:54.6...	0x00000110	Data	Standard	0x08	00 00 00 00 00 00 00 00
00000008	Receive	16:54:54.6...	0x00000110	Data	Standard	0x08	01 00 02 a8 42 05 03 3d
00000009	Receive	16:54:54.6...	0x00000110	Data	Standard	0x08	00 00 00 00 00 00 00 00
00000010	Receive	16:54:58.9...	0x00000202	Data	Standard	0x08	00 01 01 81 08 0d 01 01
00000011	Receive	16:54:58.9...	0x00000050	Data	Standard	0x08	01 08 ad 80 01 05 03 03
00000012	Receive	16:54:58.9...	0x00000050	Data	Standard	0x08	00 01 00 00 00 00 00 00

Figure 8. CAN bus analyser.

6. Conclusion

This article uses an intelligent serial port display to design a human-machine interaction testing tool for drilling instruments, defines the underground CAN bus application layer protocol, and improves the friendliness of the human-machine interaction interface. Users can directly send commands to any module connected to the CAN network through the serial port display, and the returning data can be observed, making it highly operable. Adopting the CAN bus communication method brings high real-time performance, expandability, and reliability. The serial port display has four main functions: ① reading basic information of the instrument, such as factory information and usage time; ② performing instrument angle calibration and reading the tool face; ③ sending multiple measurement control commands; ④ enabling the module to enter the self-test program. Depending on instruments' configuring and testing requirements, more controls can be added to the serial port display in the future.

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